

The Exponent Rules (Part IV) Homework

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin. Do not leave your answers with fraction exponents or negative exponents.

$$\textcircled{1} (y \cdot xy^2)^4 (-4y^5 x^{\frac{3}{2}} \cdot x)^2$$

$$\textcircled{2} (2a^{\frac{1}{5}} \cdot a^{\frac{9}{5}} b^2)^5 (-a^4 b^3 \cdot b^6)(ab)^2$$

$$\textcircled{3} (-5x^7 \cdot y^3 x^2)^3 (4cy^2)^{\frac{1}{2}} (c^0 \cdot y^3 \cdot y)^{10}$$

$$\textcircled{4} (2a^3 b^2 \cdot b^5)^2 (-6a^8 b^2 c^{\frac{1}{4}})^2 (b^4 a^{10} \cdot b^{-5} c^{\frac{1}{7}})$$

$$\textcircled{5} \left[(3y^{\frac{1}{8}} x^6)^{-3} \left(\frac{y^{\frac{1}{4}} x^{-9}}{b^0 a^2} \right)^2 \right]^2$$

$$\textcircled{6} \left[\left(\frac{3^{-2}}{3^{-4} x^7} \right)^2 \div \left(\frac{x^{-2} y^3}{10y^4} \right)^3 \right]^{-1}$$

$$\textcircled{7} \left(\frac{2a^5 y}{a^9 x^{-5}} \right)^3 \left(\frac{4b^{-3}}{y^2 b} \right)^{-2}$$

$$\textcircled{8} \left(\frac{2y^{\frac{1}{20}} a^{-1}}{5x^{-2} b^5} \right)^3 \left(\frac{a^{-8} y^3}{x^4 a^{-4}} \right)^{\frac{1}{4}}$$

$$\textcircled{9} \left(\frac{b^{-3} y^0}{a^{-2}} \right)^2 \cdot \left(\frac{b^3 a^5}{a^2 b} \right)^3 \div \frac{(2a^3 b^6)^4}{(ba^4)^{-5}}$$

$$\textcircled{10} \frac{(2x^6 y^5)^{-2}}{(3x^8 y^4)^2} \div \left(\frac{4x^5 y^4}{x^{-2} y^7} \right)^3 \cdot \left(\frac{x^0 y^{\frac{5}{6}}}{x^2} \right)^{18}$$

Answers

* Because there are multiple pathways for each problem, listing a sequence of rules along with the answers will not be helpful.

$$\textcircled{1} \frac{16y^{22}x^9}{1}$$

$$\textcircled{2} -32a^{16}b^{21}$$

$$\textcircled{3} -250x^{27}y^{50}\sqrt{c}$$

$$\textcircled{4} 144a^{32}b^{17}(\sqrt[4]{c})^9$$

$$\textcircled{5} \frac{\sqrt[4]{y}}{729x^{72}a^8}$$

$$\textcircled{6} \frac{x^8}{81,000y^3}$$

$$\textcircled{7} \frac{x^{15}y^7b^8}{2a^{12}}$$

$$\textcircled{8} \frac{8x^5(\sqrt[10]{y})^9}{125a^4b^{15}}$$

$$\textcircled{9} \frac{1}{16a^{19}b^{29}}$$

$$\textcircled{10} \frac{y^6}{2,304x^{85}}$$